

YouBeen Jeon

B.S. Candidate, Smart Vehicle Engineering, Konkuk University

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RESEARCH INTERESTS

Robot intelligence for long-horizon tasks in everyday environments: how robots can understand brief human requests, remember what they have seen and done, and plan and carry out multi-step manipulation in homes and warehouses.

- Memory-grounded task planning: semantic and spatial memory of what the robot has seen, where objects are, and which steps are done, updated as the environment and execution outcomes change, so that a high-level planner can select, chain, and revise skills over long horizons instead of relying on the current view alone
- Hierarchical planning and control: a high-level planner such as a VLM decomposes a request into skills that low-level controllers (VLA or RL policies learned from human demonstrations, or classical controllers) execute, extending from mobile manipulation to bimanual household tasks such as tidying, dishwashing, and laundry

EDUCATION

Konkuk University

B.S. in Smart Vehicle Engineering, Institute of Convergence Engineering

Seoul, Republic of Korea

Mar. 2021 – Feb. 2027 (Expected)

- Cumulative GPA: 4.06 / 4.50 Major GPA: 4.08 / 4.50

H-Mobility Class — Robotics Track

Industry-academia program: ROS/Python, kinematics & motion planning, control & dynamics, perception & state estimation

Hyundai Motor Group

Apr. 2026 – Jun. 2026

RESEARCH EXPERIENCE

Daegu Gyeongbuk Institute of Science and Technology (DGIST)

Undergraduate Research Intern, ROBIN Lab

Daegu, Republic of Korea

Jul. 2026 – Sep. 2026

- **VICON (Visual-Inertial-Contact based Hand-Object Tracking)**: Developed a multimodal human-demonstration framework that jointly captures hand motion, object pose, contact locations, and normal-force measurements using a wearable glove and a single RGB-D camera, enabling contact-rich data collection without a motion-capture setup.
 - **Occlusion-Robust Object Tracking**: designed an offline factor-graph refinement that combines visibility-aware object-pose estimates with contact-conditioned hand-motion cues, maintaining trajectories through severe hand occlusion and reducing tracking failures from 50.9% (FoundationPose) to 2.5%.
 - **Contact & Force Annotation**: reconstructed metric object meshes from monocular video and mapped synchronized glove force measurements onto the object surface to recover object-frame contact points and calibrated normal forces.
 - **Robot-Learning Data Pipeline**: built an end-to-end recording and annotation pipeline for synchronized RGB-D, hand-object trajectories, contact, and force signals, designed for force-aware manipulation and VLA training.

Korea Institute of Industrial Technology (KITECH)

Student Researcher, National Engineering R&D Center

Ansan, Republic of Korea

Jan. 2026 – Jun. 2026

- **Skill-wise Expert based Data Generation**: To reduce the high cost of teleoperation data collection, generated VLA fine-tuning data by simulator rollouts with skill-wise experts (teleop demos → Diffusion Policy based BC → Residual RL).
- **VIVA (VLM-Instructed VLA Architecture)**: Designed a hierarchical VLM-VLA system for long-horizon manipulation. Qwen3-VL decomposes commands into skill plans and asynchronously updates $\pi 0.5$ instructions from real-time scene observations. Achieved 38% / 28% / 18% success on Easy / Middle / Hard tasks, compared with 2% / 6% / 0% for the VLA-only baseline.

PUBLICATIONS & PRESENTATIONS

Submitted to IEEE International Conference on Robotics and Automation (ICRA 2027)

Sep. 2026

- **YouBeen Jeon***, UiSeong Shin*, HwanChul La*, Jaeseong Kang*, YongSeok Lee†. “VICON: Visual-Inertial-Contact based Hand-Object Tracking for Manipulation Datasets.” (*Under Review*)

KSME 2026 Spring Annual Conference

Apr. 2026

- **YouBeen Jeon***, Seunghak Bae, Youngkee Eun, Sungwook Kim. “VIVA: VLM-Instructed VLA Architecture for Long-Horizon Tasks and Skill-wise Expert based Data Generation Pipeline.” (Poster)

PROJECTS

Real-time Emergency-Stop Failure Prediction | *PyTorch, PyBullet*

Sep. 2025 – Dec. 2025

Course project (Machine Learning, Konkuk University).

- Developed a time-series classifier that predicts vehicle braking failure caused by road condition changes (rain, snow) and triggers an early warning before failure occurs.
- **PyBullet simulation environment**: collected training data by randomizing physical variables (speed, mass, friction, braking timing) and integrated the model’s threshold-based early-warning system into the simulator.
- **Transformer model training and evaluation**: achieved F1 0.89 and Acc 93.4%, selected as the final model with stable Precision of 93%, compared to GRU’s 53%.

Path Planning & Control in Complex Driving Environments | *ROS2, CARLA*

Mar. 2025 – Jun. 2025

Konkuk Dream Semester independent project.

- Built a ROS2 + CARLA autonomous-driving system that navigates safely along an optimal route to the destination across diverse urban scenarios with static and dynamic obstacles and barricaded routes, using coordinated planning and control algorithms.
- **Planning & decision-making module**: designed the ROS2 node architecture and end-to-end pipeline that smoothly switches between algorithms based on driving context using simulator sensor data, and implemented A* global path planning, cubic/B-spline interpolation with curvature-based velocity planning, RRT-based static obstacle avoidance, and D* Lite-based global replanning.

UAV-UGV Cooperative Reconnaissance Platform | *ROS2, PX4, Gazebo*

Mar. 2025 – Jun. 2025

Course project (Autonomous System Platform, Konkuk University).

- Built a cooperative-mission system for a UGV to transport a UAV to a takeoff point for reconnaissance, then meet it at a rendezvous point for landing. **Ranked 1st in course final evaluation.**
- **UGV planning and control**: designed routes and velocity profiles for the UGV to reach the takeoff point without dropping the carried UAV, and quickly return to the rendezvous point for the UAV’s precision landing.
- **UAV ArUco marker global localization**: transformed the camera-frame relative coordinates of detected markers into world-frame global coordinates using the UAV’s pose.

AWARDS & HONORS

President’s List (Rank 1 / 37) — Konkuk University

Spring 2025

- Achieved 4.50 / 4.50 GPA, KU Convergence Engineering Institute.

Seoul RISE Director's Award — 1st Physical AI Hackathon

Feb. 2026

- **Team Lead**, Team Ipalcheongchun. Track: Fine Folding (bimanual cloth folding).
- **Contribution**: Bimanual teleop setup, demo collection, ACT model training, and inference.

Encouragement Award — HL FMA 2025 aMAP Stride Championship

Nov. 2025

- **Team Lead**, Team Ganadi. MORAI + ROS1.
- **Contribution**: GPS-IMU fusion for state estimation, curvature-based velocity planning, custom ROS messages for planning and control.

Special Award — 2025 University Creative Mobility Competition, Autonomous Driving Track

Nov. 2025

- **Planning/Control SW Module Member**, Team KAI.
- **Contribution**: Centroid-graph-based Greedy path planner, control using steering sensitivity adaptive Pure Pursuit algorithm.

Excellence Award — 2025 University Creative Mobility Competition, Creative Technology Track

Nov. 2025

- **Planning/Control SW Module Member**, Team KAI.
- **Contribution**: Delaunay-triangulation based path generation for cone gate driving, control using curvature and velocity adaptive Pure Pursuit algorithm.

Grand Prize — 8th Kookmin University Autonomous Driving Competition

Aug. 2025

- **Team Lead**, Team KAI.
- **Contribution**: Reed-Shepp + Stanley parking system in Preliminary (Gazebo) / lane estimation with YOLO and obstacle distance estimation via camera-LiDAR calibration for overtaking driving in Final (ROS2, real vehicle).

ETRI Director's Award — 4th International Collegiate EV Autonomous Driving Competition

Jul. 2025

- **Team Lead**, Team S.V.E. Basic Autonomous Mobility Race.
- **Contribution**: micro-ROS based Arduino-Jetson data communication pipeline, wall-section driving using 2D LiDAR + VFH algorithm, global pose estimation via motor encoder + IMU odometry for waypoint navigation.

TECHNICAL SKILLS

Programming Languages: Python, C++

Robot Learning: LeRobot, openpi, Isaac Sim, Isaac Lab, PyTorch

Frameworks and Tools: ROS2, ROS1, Gazebo, CARLA, PyBullet, Linux, Git